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# Face-based age and gender classification using deep learning model

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## Abstract

The classification of the age and gender of face image data is a significant study subject with numerous applications. Convolution Neural Networks (CNNs) model is the effective technique to classify data, particularly for unrestricted real-world faces. In this current study, we introduce an innovative CNN-based method specifically designed for the extraction of distinguishing features from real-world facial data. These extracted features play a crucial role in the categorization and classification of images, with a focus on age and gender. We perform pre-training on facial images using the *IMDb-WIKI* dataset and subsequently classify both age and gender using unrestricted real-life facial data. To avoid over-fitting and enable the proposed model to generalise on the test images, we additionally chose a *drop-out* and data augmentation regularisation strategy. We demonstrate that optimised training hyper-parameters and well-planned network architecture produce better outcomes. The proposed technique achieved the 84.8 percent age group classification accuracy rate while the compared *CNN2ELM* model achieve the 52.3 percent accuracy rate. The suggested model demonstrates significantly better performance when contrast on the the *CNN2ELM* model on the *OIU-Adience* dataset, especially in the context of age group classification and overall accuracy.

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**Keywords:** Age group classification; Gender classification; Face image; Convolution neural networks (CNNs); Unconstrained images.

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## 1. Introduction

Due to its several common real-life approaches in fields including biometrics [1, 2], security control [3], cosmetology [4], and entertainment [5], facial recognition [6, 7] is a fascinating and pervasive topic. Particularly, the classification of faces according to age and gender has rapidly grown in popularity among other people [8]; this has a huge impact on our social life because we frequently base our interactions with others on these two features of the face [9]. The majority of widely used methods have been handcrafted, focusing on the extraction of features to investigate

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the discrimination data required to estimate the task [10, 11, 12, 13]. Numerous researchers have examined various machine learning techniques for classifying people's ages and genders, but these techniques only work well on face images taken in restrained conditions because few of them are built to manage the innumerate issues of unrestricted real-world imaging environment [14, 15]. Based on its effective feature extraction method, the Convolution Neural Networks (CNNs) model can identify the age and gender of face images [16, 17, 18, 19, 20, 21]. This demonstrates its applicability to categorise the age and gender that are not limited, automatically outperforming more traditional models [22, 23, 24, 25]. In this study, we showcase a CNN-based model tailored for the purpose of age and gender classification in real-world facial data. The main contributions of the paper as follow:

- Considering the facial data analytic work as a classification challenge, where individual age and gender categories are regarded as distinct class labels, we propose an innovative CNN-based model designed for age and gender classification within unrestricted real-life facial data.
- We recommend utilizing reliable algorithms for face detection, face landmark detection, and face alignment to enhance accuracy and performance that translate an image into the output coordinate space while detecting the face in the picture and recognising facial landmarks of unrestricted faces.
- To prevent overfitting of facial data, the suggested model is pre-trained on a sizable face ageing dataset with unrestricted age and gender labels.
- Finally, we utilise two well-known benchmark datasets for training and validation. The experiments reveal that the suggested model surpassed the compared algorithms on the same dataset when tested on *OIU-Adience* database to classified the age and gender.

The subsequent sections of this research are structured in the following manner: Section 2 explores the related literature according to age and gender categories. The recommended strategy is explained in Section 3. The findings and discussions of the benchmark datasets' unrestricted faces with age and gender labels are showcased in Section 4. The conclusion is drawn in Section 5.

## 2. Related Work

The research community has recently started to focus more on CNN, especially for its role in categorizing the age and gender of facial data obtained under unrestricted conditions [18]. Eidinger [8] categorized the age and gender of facial data acquired in challenging conditions. For the classification of age and gender in facial data collected from uncontrolled environments, Levi et al. [9] explored a five-layer CNN approach. Subhani et al. (2016) utilized direct convolutional neural system engineering to introduce a 5-layer CNN-based approach for age and gender classification using the Adience database [21]. Zhang et al. (2017) introduced a novel CNN-based model named "Residual Networks of Residual Networks (RoR)" for age and gender classification in facial data [25]. On the other hand, Duan et al. (2018) presented a hybrid age estimation model called "CNN2ELM" that predicts age and gender in facial data. This model combines three CNN models with two extreme learning machine (ELM) architectures to form CNN2ELM [26]. The suggested model produces a superior outcome. It is employed for the analysis of age and gender in real-life facial data that are not confined, making it useful for constrained images as well. In [27], several of the most recent age classification techniques are examined in detail. Early approaches [26] concentrated on the recognition of manually tuned facial features and exploited variations in these feature. While some early attempts demonstrated pretty high accuracy on limited input images, few have made an effort to raise the question brought on by differences in image quality and clarity that occur in the actual world, according to [16]. The work [2] provides a comprehensive summary of gender classification techniques. Neural networks were taken into account for gender classification as early as 1990 [17]. Support vector machines (SVMs) [10] were utilised to obtain very low error rates on gender prediction, which were of very low resolution. Kumar et al. [28] locates the features points of the face image such as nostril, mouth, eye etc., to identify the missing children using feature-based approach.

In 2015, [8] broke this norm by developing a methodology and architecture to address both age and gender. The images captured in actual environments are not exactly centred, illuminated, or aligned. Their emphasis on deep CNN is in line with a trend in the computer vision world, where CNNs are increasingly being proven to offer unmatched performance for different forms of picture classification. According to [18], the *LeNet-5* was the first CNN application.

The work [6] rekindled interest in CNNs by demonstrating that deep structures are now both practical and efficient, while the work [11] focus to achieve the good performance in recent years with the advent of previously unheard of fast and affordable processors. Because of this, the authors [8] used these developments to create a potent network that displayed cutting-edge performance. On millions of images, it demonstrates enhanced performance with deeper architecture training; the work [8] demonstrates improvements with shallower designs for their use case.

There are methods for age estimation in the literature [7], for simulating the ageing process [11], and for biometric recognition that can handle images taken at various ages [9]. Studies concentrating on broad texture features, for example, Local binary patterns (LBP) [19, 29] and Gabor features [30], fall under the first group. The research work in [23, 12] concentrate on computing fresh features particularly created for age estimation. The method described in [23] makes use of bio-inspired features built on Gabor filter pyramids. The Active Appearance Model is used in the method presented in [20] to extract the age-local feature-rich regions. The ordinal relationship of the ageing labels can provide information about learning processes, according to recent methodologies [24, 31]. In the second category, deep learning-based pattern recognition approaches have produced promising outcomes in a variety of practical applications [21, 13]. The researchers effectively created and trained Deep neural networks (DNN) to estimate the age [32, 2, 5, 25]. The majority of these methods are based on deep CNNs [5, 25]. While the other techniques, such group-aware deep feature learning [32] and *drop-out* SVMs [2], are based on various theories. Several studies [4, 10, 22] examined the viability of determining age using deep CNN-trained face recognition software. But they take time and a lot of images to perfect before being used in new situations. The suggested method, in contrast to other deep learning-based approaches, does not build or fine-tune deep neural networks but instead leverages previously learned CNNs as general feature extractors. As a result of its foundation in a simpler and quicker learning process, the proposed strategy is simpler to implement under unique circumstances. Table 1 demonstrates the findings of *OIU-Adience* database utilising classification accuracy to classify the age in the literature.

Table 1: Age group classification on *OIU-Adience* dataset using classification accuracy.

| References | Year | Method            | Age Group (%) |
|------------|------|-------------------|---------------|
| [8]        | 2014 | Dropout SVM + LBP | 45.1          |
| [9]        | 2015 | 3C2FC             | 50.7          |
| [32]       | 2016 | 3C2FC             | 54.5          |
| [27]       | 2017 | FFNN              | 58.5          |
| [23]       | 2017 | VGG               | 59.9          |
| [18]       | 2018 | CNN + ELM         | 52.3          |
| [19]       | 2018 | CNN + Focal Loss  | 54.0          |
| [26]       | 2018 | RAGN              | 66.5          |

### 3. Proposed Approach

The method for classifying age and gender within unregulated real-life facial data, as depicted in Figure 1, comprises the following fundamental elements: the activation layer employs *ReLU*, followed by a *max-pooling* layer and a local response normalization layer. These layers have consistent hyper-parameters and are positioned after the first two convolutional layers. Following the third convolutional layer is a *ReLU* activation and another *max-pooling* layer, as described in Bouchrika et al. [1]. Furthermore, all *max-pooling* layers perform computations over 3×3 regions spanning two pixels, utilizing the maximum value. To preserve resolution after the convolution operation, spatial padding is applied in both the second and third convolutional layers. After two completely linked layers, *drop-out* learning is used to lessen the chance of overfitting, according to [21]. Two fully connected layers, each consisting of 512 neurons, are followed by a *ReLU* layer and a *drop-out* layer with a *drop-out* ratio of 0.5. The 512 features obtained from the preceding layer are then mapped to 8 neurons in the final layer for age categorization. To determine a probability for each class, a *Softmax* layer with a multi-class focal loss layer is used.

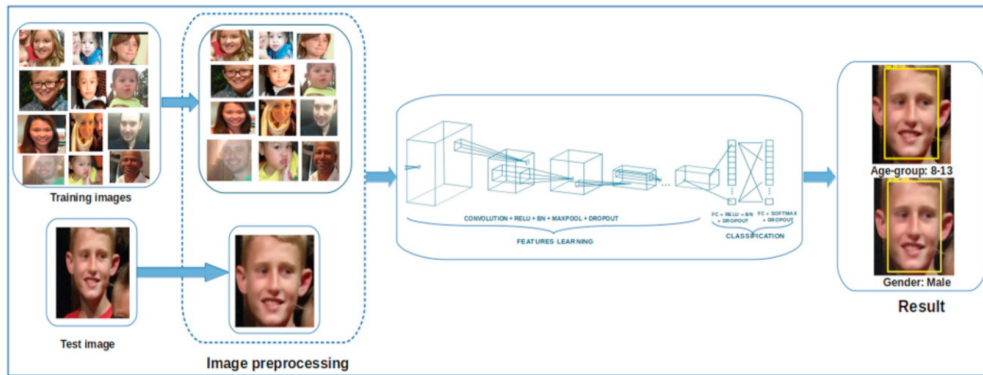


Fig. 1: Proposed Model

### 3.1. Face Detection

In the initial stage, face detection is performed using the 'pre-trained HOG (Histogram of Oriented Gradients) + Linear SVM' library available in the *dlib* library [31].

Linear SVM implements an original proprietary version of benchmark approach to create the linear SVM, is the most recent highly quick machine learning algorithm for resolving multi-class classification issues from extremely huge data sets [29]. The most recent, lightning-fast machine learning algorithm, Linear SVM [29] solve multi-class classification problems from exceedingly large data sets.

### 3.2. Facial Landmark Detection and Alignment

During the face detection process, the face landmark method described in [10] is applied to locate key facial components such as the mouth, right and left eyebrows, right and left eyes, nose, and jaw in the facial image. Commonly used landmarks include the corners of the eyes, the nose, nostrils, the mouth, and the arcs of the eyebrows. The proposed landmark detection technique swiftly identifies these facial landmarks on unrestricted faces. The CNN model comprises four layers: convolutional, ReLU activation, pooling, and a fully connected layer.

#### 3.2.1. Convolutional

Since it is impossible to predict where these characteristics will appear in a new image, CNN looks for them everywhere and in all potential contexts. To assess how well a feature matches a section of the image. We perform pixel-wise multiplication between the feature and the corresponding pixels in the image during the initial stage. We then subtract the sum of these responses from the total number of pixels in the feature. For instance, when both pixels are white, the result is 1 ( $1 * 1$ ), and if both are black, it's equivalent to  $(-1) * (-1)$ . Each matched pixel earns 1 in each scenario. Any difference additionally costs  $-1$ . When a feature's pixels completely match, its sum is 1, and its number is divided by the feature's total. Likewise, if none of the pixels in a feature correspond to the image patch, the response is  $-1$ . We continue this procedure to finish the convolution by matching the feature with each feasible image patch. Determined by the position of each patch in the image, we may use the output of each convolution to create a new 2D array. The final image was filtered to create this map of matches. The location of the feature within the image is shown on a map. Values close to 1 indicate strong matches, values near  $-1$  suggest matches for the image with the opposite attributes, and values near 0 signify no matches of any kind.

#### 3.2.2. Pooling

Pooling is a method for compressing large images while keeping the most important details. The concept behind pooling is quite simple, akin to that of a second-grader. As a small window slides across an image, we simply pick the maximum value within that window each time. After pooling, an image contains about one-fourth less pixels than it did before. We keep the best match for each feature within the window because it keeps the most value from each

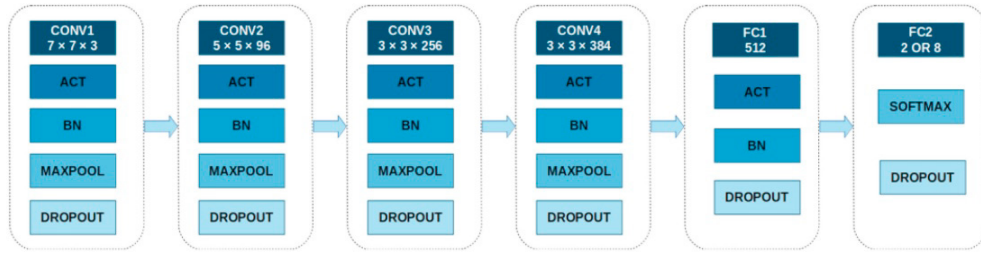


Fig. 2: Proposed CNN Model

window. It does not particularly care where the feature fits within the window, as long as it does so. This aids in resolving the issue of computers becoming overly literal.

### 3.2.3. Rectified Linear Units (ReLU)

Every time a negative integer appears, *ReLU* logic substitutes a 0. By preventing obtained values from becoming close to 0 or towards infinity, this keeps CNN logically sound. A *ReLU* layer outputs data that has a size equal to the input's but is empty of negative values.

### 3.2.4. Fully connected layers

Through fully connected layers, the high-level filtered images are converted into votes. We simply need to choose from the two categories *X* and *O* in this situation. Traditional neural networks are mostly composed of fully connected layers. Inputs are treated as a single list rather than a 2D array, and each value is given equal consideration. Each value casts its own vote to determine whether the current image is an *X* or an *O*. These receive more votes than the other options. The results of these votes are represented as weights, or the degree to which each value and each category are connected. A new image that is delivered to CNN gradually ascends via the lower layers until it reaches the top, fully connected layer. After that, a vote is held. The input's category is determined by which response receives the most votes. Affine transformation is a linear mapping technique that keeps planes, straight lines, and points in tact [33].

## 3.3. Architecture of proposed CNN Model

We depict the architecture of the proposed model, which is depicted in Figure 2. Both feature extraction and categorization are steps in this process. The feature extraction phase consists of four convolutional layers, each with specific parameters such as the number of filters, filter size, and stride. The first convolutional layer utilizes 96 filters of size 7×7 with a stride of 4×4. The second convolutional layer employs 256 filters of size 5×5. The third layer is similar to the second but has 384 filters of size 3×3 instead of 5×5. The fourth and final convolutional layers use 256 filters of size 3×3. All convolutional layers include a fixed *drop-out* of 25% to enhance generalization and mitigate overfitting.

During the classification phase, age group and gender tasks are categorized using two fully connected layers. The first fully connected layer consists of 512 neurons and is followed by a batch normalization layer, a *ReLU* activation layer, and a *drop-out* layer with a 50% *drop-out* ratio. The last fully connected layer maps 512 features to eight or two neurons, depending on the classification task. Probabilities for each class are computed using *Softmax* with a cross-entropy loss function.

**Cross-Entropy:** Cross-entropy loss produces an output that ranges from 0 to 1 and gauges the effectiveness of a classification model. In binary classification, *N* equals 2, meaning there are two classes [Eq. 1].

$$-(z \log(p) + (1 - z) \log(1 - p)) \quad [29] \quad (1)$$

In multi-class classification when  $N > 2$ , we compute individual losses for each label and then aggregate them to obtain the overall loss (Eq. 2).

$$-\sum_{c=1}^N z_{o,c} (\log p_{o,c}) \quad [29] \quad (2)$$

where

- $N$  denotes the number of classes.
- If the class label  $c$  corresponds to the correct classification for observation  $o$ , then  $z$  serves as a binary indicator, taking on values of either 0 or 1.
- $p$  represents the predicted probability of the observation belonging to class label  $c$ .

#### 4. Results and Discussion

The training details for both classifiers, which effectively estimate the age group and gender of uncontrolled facial data, are provided for the *Adience* database. While the gender classifier divides gender into two classes, the age classifier will be in charge of forecasting the ages of eight different classes. We perform pre-training on the two CNN-based classifiers using the *IMDb-WIKI* database, which comprises uncontrolled real-world facial data along with age and gender descriptors. We partitioned the *IMDb-WIKI* dataset into two categories: 90% are used for training, 10% are used for validation, and 70% of the *OIU-Adience* images are shared equally between testing and validation to make up the other 30%. Before being sent into the network, the images in the database were first cropped to 224×224 pixel and then re-scaled to 256×256 pixel.

Additionally, we train the network with 64-batch sizes. The proposed model for the classifiers is optimized using *stochastic* gradient descent with mini-batches of 256, a *momentum* value of 0.9, and a *weighted decay* of 0.0005. In case there is no improvement in accuracy, the initial learning rate is configured as 0.0001 and subsequently reduced by a factor of 10. To further enhance the efficiency of the proposed model, we apply data augmentation to both the training and testing images, along with *drop-out* regularization techniques. We compute *SGD* as described in Equation 3:

$$\beta = \beta - \eta \cdot \nabla \beta J(\beta; x^{(i)}; y^{(i)}) \quad (3)$$

where

- $\eta$  represents the learning rate.
- $\nabla \beta J$ , the gradient of the loss term concerning the weight vector  $\beta$ .

The *OIU-Adience* and *IMDb-WIKI* databases, along with experimental testing of the proposed model on *OIU-Adience* using age labels, are described in depth.

##### 4.1. Description of the Dataset

Initially, we train the proposed model using the *IMDb-WIKI* database [25], and subsequently fine-tune it using the original *OIU-Adience* database containing unconstrained facial data [8]. The *OIU-Adience* dataset [25] consists of approximately 26,000 face images obtained under ideal real-life and unrestricted conditions. It contains two gender labels and eight different age ranges (0-2, 4-6, 8-13, 15-20, 25-32, 38-43, 48-53, and 60+). The *IMDb-WIKI* database

[25] contains over 500,000 images with precise age labels spanning from 0 to 100 years. Table 2 displays the specifics of both datasets.

Table 2: Description of the face ageing datasets.

| Database               | Size   | Subject | Age group | Age Type  | Year | In-the-wild |
|------------------------|--------|---------|-----------|-----------|------|-------------|
| <i>OIU-Adience</i> [8] | 26580  | 2284    | 0-60+     | Age Group | 2014 | Yes         |
| <i>IMDb-WIKI</i> [25]  | 523051 | 20284   | 0-100     | Real      | 2016 | Yes         |

Unconstrained face images can now be categorised by age and gender using a unique CNN model. The efficiency of the suggested technique to classify a person into the proper age group in the *Adience* database has been examined in a number of empirical experiments. Two widely used metrics to assess the effectiveness of the two classifiers in the literature are accuracy and confusion matrices.

**Confusion Matrix** [29, 34]: The effectiveness of the classification model on a test dataset is assessed through a confusion matrix table. This metric gauges the effectiveness of the classification method by considering four distinct combinations of expected and actual class outcomes. The binary gender classification results and the eight age groups (0-2, 4-6, 8-13, 15-20, 25-32, 38-43, 48-53, 60+) classification results are presented using respective confusion matrices. These findings are generated by applying the proposed approach to the *OIU-Adience* database for age group and gender categorization.

**Accuracy:** This determines how closely the measured value relates to the reference or established value. The face data are correctly classified into age (or gender) categories. Out of all the face data samples analysed, it determines the percentage of true results—both real positive and real negative (Eq. 4).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad [31] \quad (4)$$

where

- The true positive number is abbreviated as TP.
- TN denotes the count of true negative values.
- FP signifies the false positive values.
- FN stands for the false negative values.

The classified results for age variation are shown in the Figure 3a and Figure 3b. From the confusion matrix table in the Figure 4a and Figure 4b, as contrast to the other age groups, it can be demonstrated that the age group labels (8-13) and (0-2) are predicted with the highest accuracy. For (0-2) age group range, the infant face images have distinguishing characteristics that make it simple for the classifier to identify this age range. That might be due to its size and distinguishing qualities in those image categories for (8-13) groups. Due to its limited sample size, the age group labels (48-53) exhibited the lowest level of precision. We employ a confusion matrix measure on the *OIU-Adience* dataset to assess the accuracy of the optimal configuration of the proposed method in relation to classification accuracy. We subsequently contrast these outcomes with those accomplished by the *CNN2ELM* model. The suggested model achieves an age group classification accuracy of 84.8 percent, which is 18.3 percent higher than the best-reported benchmark results of *CNN2ELM* model in Duan et al. [26]. Therefore, the proposed method obtain the 78.42 percent more accuracy rate as compared to the *CNN2ELM* model on the benchmark database. Therefore, the proposed model shows the better results because it is more efficient and fast classification ability than the compared model.

As shown in Figure 5 and Figure 6, the proposed method has been shown to accurately estimate the age range of faces. However, there are a small number of instances where face image were misclassified. This could be due to

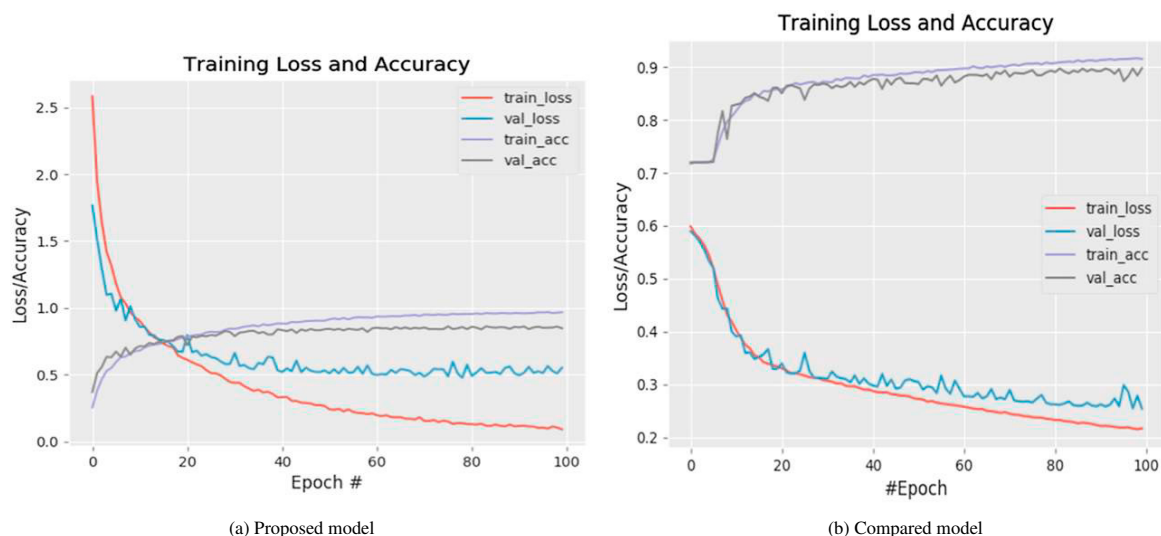


Fig. 3: Graph of accuracy results for age group classification

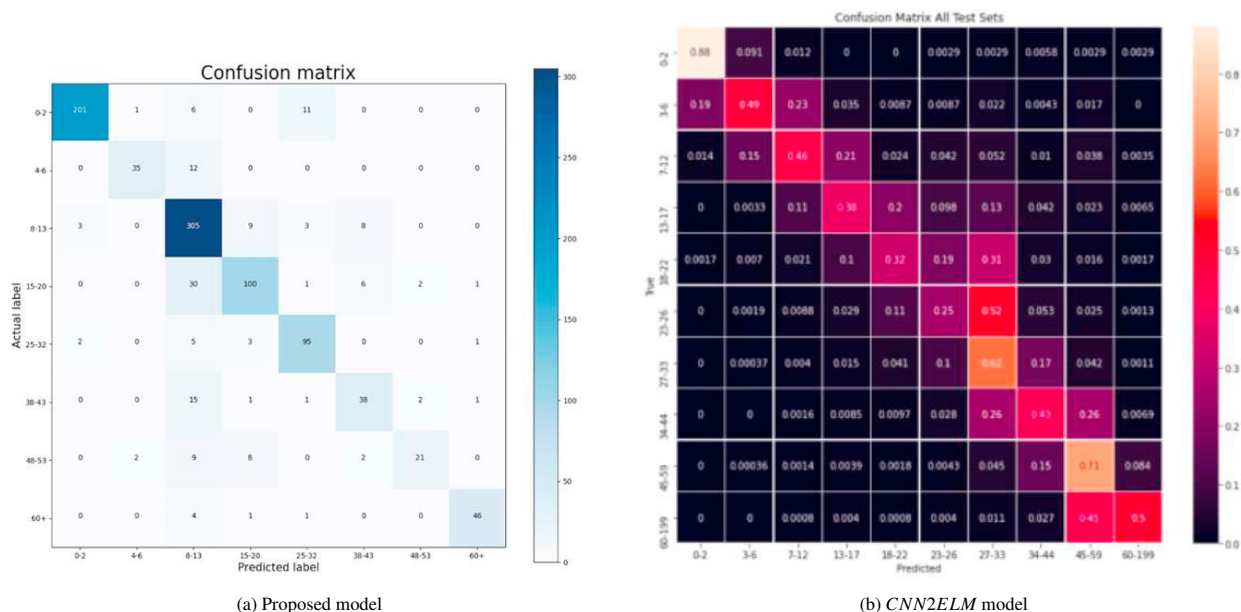


Fig. 4: Graph of confusion matrix results for age group classification

various degrees of variability (lack of consistency or fixed pattern) associated with unconstrained images, such as low resolution, non-frontal, lighting circumstances (Figure 7 and Figure 8).

The proposed technique has an age group categorization accuracy rate of 84.8 percent. However, the *CNN2ELM* model achieves classification accuracy values for the age group of 52.3 percent. The proposed model achieved a classification accuracy that was 38.33 percent higher than that of the *CNN2ELM* model on the benchmark database. This is as shown in Figure 9a. The proposed technique obtains mean absolute error values for age group categorization of 2.26. Conversely, the *CNN2ELM* model attains a Mean Absolute Error (MAE) value of 5.55 for age group classification. Therefore, the proposed model obtained the 59.29 percent lower MAE value than the compared model on the benchmark database. This is as shown in Figure 9b.

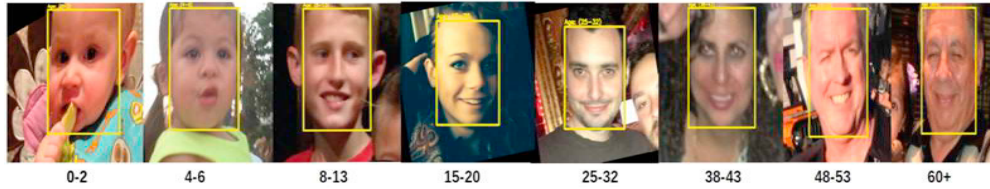
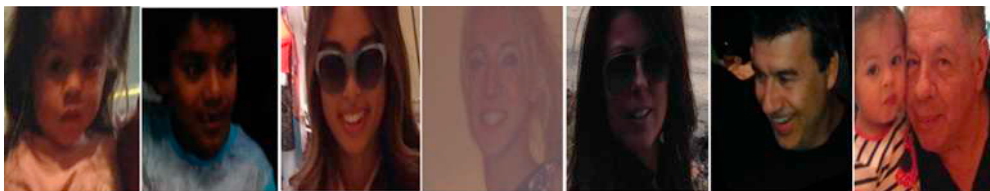


Fig. 5: Age group classification (Proposed model)

Fig. 6: Age group classification (*CNN2ELM* model)

Fig. 7: Faces with misclassification (Proposed model)

Fig. 8: Faces with misclassification (*CNN2ELM* model)

## 5. Conclusion

The suggested CNN-based model categorizes both the age and gender of uncontrolled real-world facial data. The proposed method depended on CNN architecture's skill at feature extraction and classification. The proposed model is initially pre-trained using the *IMDb-WIKI* database and subsequently fine-tuned with the original database. It is primarily responsible for the classification model's good performance. The classification accuracy of the approach is significantly improved by employing a robust face detection and alignment technique. The new model's suitability for classifying people by age and gender using the *OIU-Adience* database has been thoroughly evaluated, which supports the method's viability for use with unrestricted real-life face data. It has been shown that the proposed model demonstrates advancements over the *CNN2ELM* model in term of age classification and accuracy rate. The proposed model achieve 84.8 percent age group classification accuracy rate as compared to *CNN2ELM* model that obtained 52.3 percent accuracy rate on the *OIU-Adience* dataset. The proposed model achieves a mean absolute value of 2.26 for age group classification, in contrast to the mean absolute value of 5.55 for the same age group obtained by the *CNN2ELM* model. Hence, the proposed model attains a superior accuracy rate compared to the *CNN2ELM* model in both age group classification and overall accuracy. In the subsequent times, our intentions are to concentrate on processing facial image data that include various facial poses, such as tilted faces.

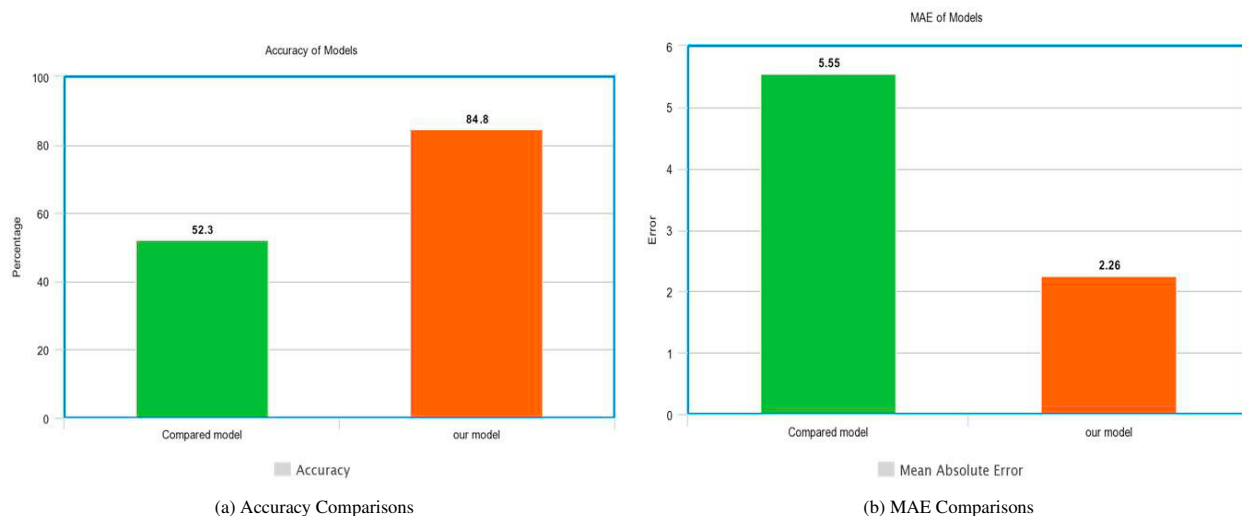


Fig. 9: Accuracy and MAE comparisons of both models

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